Pet milk replacers & specialty liquids: HMTc Standard

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CORRESPONDENCE

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KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

This revision 2026-08.

Abstract

The Heavy Metal Tested & Certified human-grade standard for pet milk replacers & specialty liquids, stated as consumed. Pet food is legally feed; HMTc voluntarily certifies each analyte to the strictest human-food sovereign maximum for the nearest analog, with the laxer feed-law floor disclosed for contrast.

1 | How every number on this standard is set

Pet food is legally governed by feed law — EU Directive 2002/32/EC on undesirable substances in animal feed, FDA Center for Veterinary Medicine guidance, and AAFCO model regulations — whose maximum levels are generally laxer than the corresponding human-food limits. Japan (FAMIC pet-food standard) and China (draft pet-food guidance) set enforced or proposed pet-food maxima in the same lax range. The HMTc standard for each analyte is not the feed limit: it is the strictest applicable human-food sovereign maximum for the nearest human-food analog, adopted as a voluntary human-grade over-compliance. The feed floor is disclosed on every row and never selects the number.

2 | Master limit table

All values in ppb (µg/kg), as consumed basis. A brand meeting every value qualifies for the mark.

ANALYTE	HMTC LIMIT	BASIS

Lead (Pb)	20	read-across (India)
Cadmium (Cd)	50	read-across (Paraguay)
Total arsenic (tAs)	50	read-across (Paraguay)
Inorganic arsenic (iAs)	≤ tAs	bounded by tAs (reflex speciation)
Mercury, total (tHg)	10	read-across (Hong Kong)
Methylmercury (MeHg)	≤ tHg	bounded by tHg (reflex speciation)
Nickel (Ni)	100	read-across (EU 2024/1987)
Tin (Sn)	250,000	read-across (Singapore)
Aluminium (Al)	500	read-across (Australia / New Zealand)
Total chromium (Cr)	300	read-across (GB 2762)
Hexavalent chromium (Cr-VI)	≤ Cr	bounded by Cr (reflex speciation)

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION
Lead (Pb)	20	read-across (India); Secondary milk products (as consumed)
Cadmium (Cd)	50	read-across (Paraguay); fluid milk ready for consumption and dairy products without addition, undiluted and unconcentrated
Total arsenic (tAs)	50	read-across (Paraguay); fluid milk ready for consumption and dairy products without addition, undiluted and unconcentrated
Inorganic arsenic (iAs)	≤ tAs	bounded by tAs (reflex speciation)
Mercury, total (tHg)	10	read-across (Hong Kong); Secondary milk products
Methylmercury (MeHg)	≤ tHg	bounded by tHg (reflex speciation)
Nickel (Ni)	100	read-across (EU 2024/1987); infant/young-child formula, liquid
Tin (Sn)	250,000	read-across (Singapore); any food

Aluminium (Al)	500	read-across (Australia / New Zealand); infant formula, follow-on formula, and special medical purpose product for infants other than preterm-infant products
Total chromium (Cr)	300	read-across (GB 2762); raw, pasteurized, sterilized, modified and fermented milk (GB 2762-2022 Table 7, section 4.7); total Cr, not Cr-VI
Hexavalent chromium (Cr-VI)	≤ Cr	bounded by Cr (reflex speciation)

4 | Cite this standard

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References

[1] India
Binds Pb.
READ-ACROSS

[2] Paraguay
Binds Cd, tAs.
READ-ACROSS

[3] Hong Kong
Binds tHg.
READ-ACROSS

[4] EU 2024/1987
Binds Ni.
READ-ACROSS

[5] GB 2762
Binds Cr.
READ-ACROSS

[6] Singapore
Binds Sn. any food.
READ-ACROSS

[7] Australia / New Zealand
Binds Al.
READ-ACROSS

STATUS AND LICENCE COMPETING INTERESTS

A normative certification standard, editorially reviewed and version-controlled in-house; every value traces to a named sovereign maximum level, disclosed in the derivation. © 2026 Institute of Contaminant Standards / Heavy Metal Certified. All rights reserved.

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DATA AVAILABILITY

The works this standard draws on are listed in full in the references above and on the standard page at https://heavymetalcertified.com/standards/pet-milk-replacers-specialty-liquids. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

PUBLICATION HISTORY

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/pet-milk-replacers-specialty-liquids.